

LIBERO-Plus: A Progressive Robustness Benchmark for Visual-Language-Action Models

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Abstract

Visual–Language–Action (VLA) models report impressive success rates exceeding 95% on robotic manipulation benchmarks, yet these results may mask fundamental weaknesses in robustness. Current simulation-based robustness evaluations suffer from narrow perturbation coverage, manual design constraints, and coarse-grained analysis that fails to reveal when and how models fail. To address this gap, we propose LIBERO-Plus, a comprehensive, automatic, and fine-grained evaluation framework with controlled perturbations across seven dimensions: object layouts, camera viewpoints, robot initial states, language instructions, lighting conditions, background textures, and sensor noise. Our systematic analysis of ten state-of-the-art models reveals consistent brittleness beneath apparent competence, with performance dropping from 95% to below 30% under modest perturbations. Our findings challenge the assumption that high benchmark scores equate to true competency and highlight the need for evaluation practices that assess reliability under realistic variation.

1. Introduction

Recent advances in Visual–Language–Action (VLA) models have achieved remarkable performance on embodied manipulation tasks [2, 5, 9, 11, 12, 14, 20, 22], with systems reporting success rates exceeding 95–99% on popular simulation benchmarks like LIBERO [16]. While these results suggest that manipulation might be close to being “solved” under current evaluation protocols, real-world deployment often reveals a stark contrast, with models struggling when faced with environmental variations such as lighting changes, shifts in camera viewpoints, or different

natural language phrasings. While simulation environments provide essential controlled settings for fair model comparison and reproducible evaluation rather than costly real-world experimentation, current simulation-based evaluation practices focus primarily on performance under fixed, ideal conditions and fail to assess model robustness across the diverse variations that comprehensive simulation environments can systematically provide.

Existing simulation-based efforts to address robustness evaluation have introduced perturbations along specific axes, such as object variations, lighting changes, and paraphrased instructions [6, 7, 17, 21, 23, 26]. While valuable, these simulation-based initiatives have limitations in three key areas. First, they often focus on narrow perturbation types rather than providing a systematic evaluation across a broad spectrum of robustness factors that simulation environments can controllably generate. Second, many rely on manually designed perturbations, which constrain scalability and reproducibility. Third, most evaluations report aggregate success rates, which mask the nuanced failures and make it difficult to pinpoint the exact conditions under which a model fails. As a result, current simulation-based approaches offer fragmented insights but lack the methodological rigor to systematically characterize the boundaries of model capabilities or guide targeted improvements.

In this work, we introduce LIBERO-Plus, a systematic framework for progressive robustness evaluation of end-to-end VLA models. Building on the widely adopted LIBERO benchmark, our framework extends evaluation along three methodological directions. First, we provide comprehensive perturbation coverage across seven major categories and fourteen subcategories, including camera viewpoints, robot initial states, language instructions, lighting conditions, background textures, sensor noise, and object layouts, moving beyond narrow perturbation types to systematic evaluation across a broad spectrum of robustness factors. Second, we enable automated and scalable genera-

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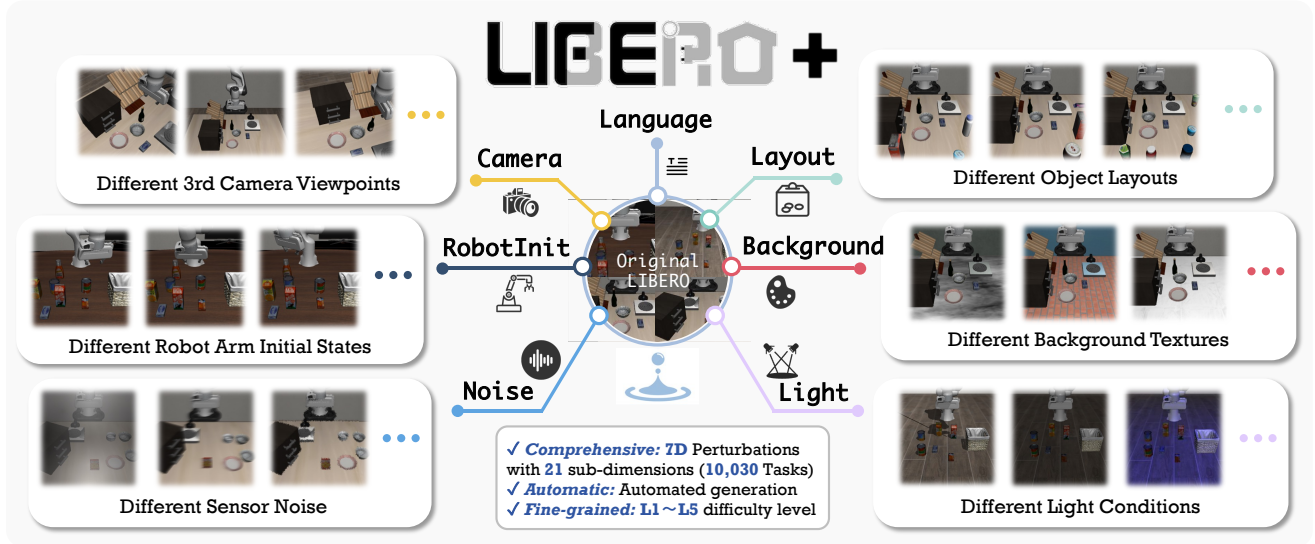


Figure 1. Overview of LIBERO-Plus. Building upon the original LIBERO, we systematically generate perturbations along 7 controllable dimensions—camera viewpoint, robot initial state, language instruction, lighting, background texture, sensor noise, and object layout. Each dimension is automatically parameterized to produce a diverse set of sub-settings, systematically calibrated into 5 difficulty levels (L1–L5), enabling large-scale, reproducible, and fine-grained robustness evaluation.

tion through parameterized methods that can construct both training and testing datasets at scale, encompassing over 56K robustness scenarios and ensuring reproducible evaluation without manual design constraints. Third, we introduce fine-grained analysis through progressive difficulty stratification (L1–L5), moving beyond aggregate success rates to capture detailed robustness profiles that reveal exactly when and how models fail under different perturbation intensities.

Through the application of LIBERO-Plus, we analyze the robustness of ten VLA models, revealing that current models overfit to specific conditions, often showing significant performance degradation when exposed to moderate or challenging perturbations. Our study also highlights several core weaknesses in contemporary VLA models: **Vulnerability to Visual Shifts:** an over-reliance on fixed visual features leads to failure under variations in camera angle or illumination; **Inadequate Kinematic Reasoning:** limited generalization across different initial robot configurations reflects a lack of deep kinematic understanding; **Superficial Language Interaction:** linguistic inputs are often underutilized or even completely ignored, as shown by the minimal impact of instruction variation.

Our main contributions are as follows:

1. A novel benchmark, LIBERO-Plus, offering comprehensive, fine-grained, and automated evaluation of model robustness under a wide array of perturbations.
2. A thorough evaluation of 10 popular VLA models, highlighting the overfitting and lack of generalization exhibited by models under existing simulation benchmarks.
3. Detailed analysis that provides richer insights into the

robustness landscape, offering critical insights for improving model development and robustness.

2. Related Work

2.1. Vision-Language-Action Models

Recent advancements in Vision-Language-Action (VLA) models have expanded the paradigm of foundation models from language and vision into robotics, motivating unified architectures. Autoregressive approaches [3, 11, 14, 15, 20, 24] discretize robot actions into tokens and train end-to-end policies on large-scale demonstrations, while diffusion-based models [1, 2, 13, 25] generate continuous trajectories via generative diffusion experts. More recently, reinforcement learning methods [8, 17, 18, 22] move beyond supervised fine-tuning, emphasizing robustness and downstream adaptability through reinforcement learning objectives. However, while these models show “zero-shot” performance in familiar settings, their success often reflects interpolation rather than true generalization. As a result, existing benchmarks lack comprehensive insights into model performance under distribution shifts, highlighting the need for systematic and fine-grained robustness evaluations.

2.2. Generalization Robotic Manipulation Evaluations

Evaluating the generalization ability of robotic manipulation models has remained a significant challenge. Early efforts [10, 16, 19] established reproducible environments for model testing. However, these benchmarks primarily

Table 1. Comparison of different simulated generalization evaluation benchmarks for VLA models.

Method	Automation	Simulator	Fine-grained	Perturbation Dimensions						
				Layout	Background	Light	Camera	Robot	Language	Noise
AGNOSTOS [26]	✗	RLBench	✗	✓	✗	✗	✗	✗	✗	✗
RL4VLA [17]	✗	ManiSkill	✗	✓	✓	✗	✗	✓	✓	✗
INT-ACT [6]	✗	ManiSkill	✗	✓	✗	✗	✗	✗	✓	✗
Gembench [7]	✗	RLBench	✗	✓	✗	✗	✗	✗	✗	✗
VLATest [23]	✓	ManiSkill	✗	✓	✗	✓	✓	✗	✓	✗
COLOSSEUM [21]	✓	RLBench	✗	✓	✓	✓	✓	✗	✗	✗
LIBERO-Plus (Ours)	✓	LIBERO	✓	✓	✓	✓	✓	✓	✓	✓

focused on specific tasks, with a limited scope for evaluating robustness across diverse conditions. Some studies [6, 7, 17, 26], relied on manually designed tasks to cover several perturbation dimensions. These studies typically used small sample sizes, often with fewer than 100 test scenarios. This limited the ability to perform large-scale, systematic evaluations of model robustness across a broad range of conditions. Other benchmarks [21, 23], addressed this issue by automating task generation. These frameworks significantly increased the number of tasks, enabling broader coverage of perturbation dimensions, including lighting and camera viewpoints. However, these automated approaches still lack a fine-grained analysis within each perturbation dimension, limiting the depth of insights into model performance under varying conditions. In contrast, our LIBERO-plus offers three key advancements: (1) **Comprehensive**: Evaluation across seven robustness dimensions and 21 sub-dimensions, ensuring a thorough assessment of model generalization. (2) **Automation**: Automated task generation, enabling the creation of a large number of diverse training and test samples. (3) **Fine-grained**: Incorporation of difficulty levels (L1–L5), providing detailed insights into when and how VLA models fail.

3. LIBERO-Plus

LIBERO-Plus is a comprehensive benchmark for systematically evaluating VLA model generalization in robotic manipulation. It extends LIBERO with 10,030 task instances across 7 perturbation factors and 21 sub-dimensions.

3.1. Benchmark Construction

We build on LIBERO for its diverse manipulation scenarios with language instructions and multi-view observations. Perturbation factors are environmental and task properties systematically varied to induce distribution shifts. These include: (1) *Objects Layout*: introduction of confounding objects and spatial displacement of target objects; (2) *Camera Viewpoints*: modifications to camera pose, orientation, and field-of-view; (3) *Robot Initial States*: variation

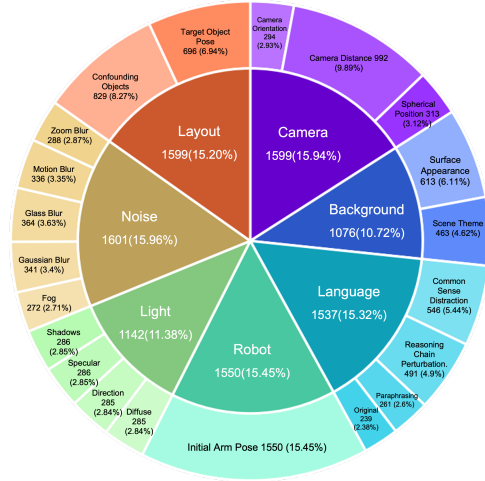


Figure 2. Architecture of the LIBERO-Plus benchmark, comprising 10,030 tasks organized across seven perturbation factors and twenty-one underlying components.

in manipulator starting poses; (4) *Language Instructions*: semantic rewrites with increased linguistic complexity; (5) *Light Conditions*: variations in illumination intensity, direction, and color; (6) *Background Textures*: alterations to surface materials and textures; and (7) *Sensor Noise*: photometric distortions including jitter and Gaussian blur. These perturbations create scenarios where the input distribution changes while the underlying task structure remains invariant, representing covariate shift in out-of-distribution generalization. Full per-factor specifications and construction details are provided in Appendix A.

3.2. Benchmark Statistics

Figure 2 shows our benchmark’s hierarchy across perturbation factors and their components. LIBERO-Plus is built by systematically applying seven perturbation factors to original LIBERO tasks, then stratifying them into five difficulty levels (Level-1 to Level-5) based on empirical performance. This stratification is derived from evaluating tasks with four

representative VLA models, creating a graduated framework that captures varying degrees of generalization challenge, with level-wise statistics in Appendix D.

3.3. Evaluated Methods

We analyze ten representative VLA models spanning diverse architectural paradigms and training methodologies: (1) OpenVLA [11] and its variants (2) OpenVLA-OFT [12], (3) OpenVLA-OFT_w (third-view-only version), (4) OpenVLA-OFT_m (mix-sft version, trained on all 4 suites), (5) π_0 [2], (6) π_0 -fast [20], (7) Nora [9], (8) WorldVLA [5], (9) UniVLA [4] and (10) RIPT-VLA [3]. These models represent diverse training paradigms including web-data co-training, world modeling, and reinforcement learning. Detailed model specifications and implementation details are provided in Appendix B.

4. How Do Single-Dimension Perturbations Affect VLA Models?

We present the main experimental results in Table 2 and Figure 3, which collectively reveal a significant fragility in the generalization capabilities of current VLAs. As shown, even minor perturbations can lead to drastic performance degradation. Below we analyze the specific robustness patterns across perturbation dimensions, models, and tasks.

Finding 1: Significant Overall Fragility to Perturbations. Across all perturbation factors, current VLAs exhibit brittle generalization. Performance degrades significantly under various input perturbations, particularly with changes in camera viewpoint and robot initial state.

Finding 2: Robustness varies considerably by perturbation type. Models are most vulnerable to changes in camera viewpoint and robot initial state, which require a high-level understanding of spatial geometry and proprioception. In contrast, they show relative resilience to lighting and background variations, which constitute more superficial, low-level visual changes.

Finding 3: Minor Impact of Language Perturbation. Contrary to expectations, language perturbations result in the second smallest performance drop (-25.3) on average, a counter-intuitive result that warrants further investigation. As discussed in Section 6, this likely does not stem from better linguistic generalization. Instead, as empirically validated, a more plausible explanation is that models may rely less on language instructions than expected, possibly by extracting task cues from visual context.

Finding 4: Model robustness is dictated by architecture and training paradigm. Specifically, models incorporating a first-person wrist camera (e.g., OpenVLA-OFT) demonstrate superior generalization, especially to camera viewpoint changes, compared to those reliant solely on a third-person view (e.g., OpenVLA-OFT_w). Furthermore, training strategies that emphasize diversity and co-training

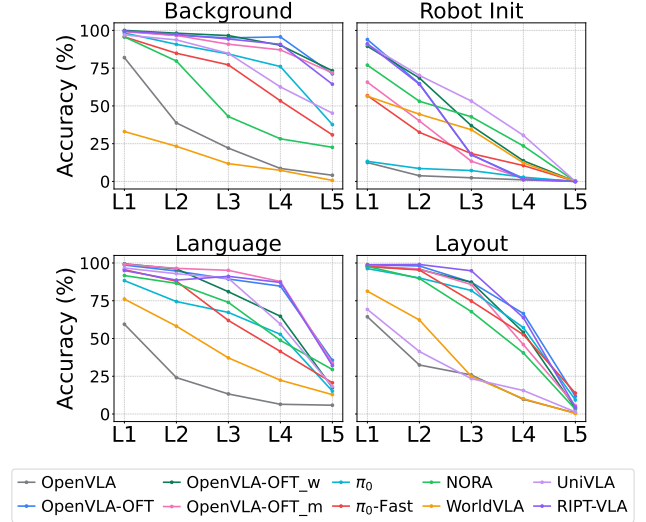


Figure 3. Model performance trends across perturbation difficulty levels. The line plots show the success rate of each model as the intensity of four different perturbation dimensions increases.

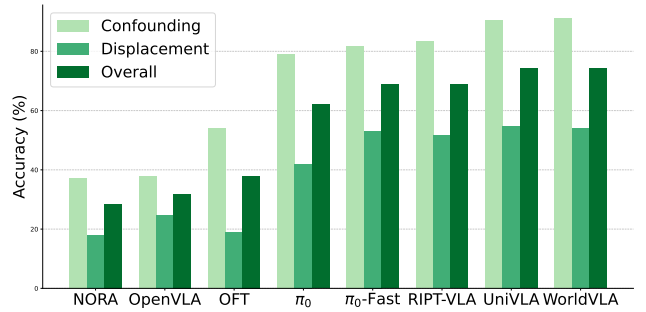


Figure 4. Robustness to object layout perturbations. Comparison of different models under confounding and displacement perturbations, as well as their overall robustness.

(e.g., π_0 , π_0 -fast) consistently yield more robust models across multiple perturbation types, highlighting the importance of exposure to varied data distributions.

5. Do contemporary VLA Models truly pay attention to visual inputs?

While the overall trends reveal substantial fragility, we observe two particularly interesting patterns in the data: (1) models exhibit surprising resilience to background changes, and (2) several models show limited sensitivity to light variations. These observations raise important questions about what representations the models are actually learning. Do they genuinely understand task-relevant object semantics, or are they relying on superficial visual cues? To answer these questions, we conduct finer-grained analyses of object layout and illumination robustness.

Table 2. Model performance under different perturbations. For each model, the first row reports task success rate (%) per perturbation dimension, with the “Original” column indicating the performance on unperturbed inputs. The second row (denoted by ↓) shows the corresponding absolute performance drop. The results highlight significant variations in robustness across models and perturbation types.

	Original	Camera	Robot	Language	Light	Background	Noise	Layout	Total
OpenVLA	76.5	0.8	3.5	23.0	8.1	34.8	15.2	28.5	15.6
		↓75.8	↓73.0	↓53.5	↓68.4	↓41.7	↓61.3	↓48.0	↓60.9
OpenVLA-OFT	97.1	56.4	31.9	79.5	88.7	93.3	75.8	74.2	69.6
		↓40.8	↓65.2	↓17.6	↓8.4	↓3.8	↓21.3	↓22.9	↓27.5
OpenVLA-OFT_w	95.3	10.4	38.7	70.5	76.8	93.6	49.9	69.9	55.8
		↓84.9	↓56.7	↓24.8	↓18.5	↓1.7	↓45.4	↓25.4	↓39.5
OpenVLA-OFT_m	97.6	55.6	21.7	81.0	92.7	91.0	78.6	68.7	67.9
		↓42.0	↓75.9	↓16.6	↓4.9	↓6.6	↓19.0	↓28.9	↓29.7
π_0	94.2	13.8	6.0	58.8	85.0	81.4	79.0	68.9	53.6
		↓80.4	↓88.2	↓35.4	↓9.2	↓12.8	↓15.2	↓25.4	↓40.6
π_0 -fast	85.5	65.1	21.6	61.0	73.2	73.2	74.4	68.8	61.6
		↓20.4	↓63.9	↓24.5	↓12.3	↓12.3	↓11.1	↓16.7	↓23.9
Nora	87.9	2.2	37.0	65.1	45.7	58.6	12.8	62.1	39.0
		↓85.7	↓50.9	↓22.8	↓42.2	↓29.3	↓75.1	↓25.8	↓48.9
WorldVLA	79.1	0.1	27.9	41.6	43.7	17.1	10.9	38.0	25.0
		↓79.0	↓51.2	↓37.5	↓35.4	↓62.0	↓68.2	↓41.1	↓54.1
UniVLA	95.2	1.8	46.2	69.6	69.0	81.0	21.2	31.9	42.9
		↓93.5	↓49.0	↓25.7	↓26.2	↓14.3	↓74.0	↓63.3	↓52.3
RIPT-VLA	97.5	55.2	31.2	77.6	88.4	91.6	73.5	74.2	68.4
		↓42.3	↓66.3	↓20.0	↓9.2	↓5.9	↓24.0	↓23.3	↓29.1

Do Models Genuinely Attend to Task-Relevant Objects?

We are pleasantly surprised to observe that the models are relatively insensitive to changes in the *Background* setting. To further investigate whether the models truly focus on the core interactive objects and genuinely understand the high-level semantics and spatial information relevant to the task, we decomposed the *Object Layout* perturbation into two subcategories: (1) *adding confounding objects*, and (2) *changing the placement and pose of the target objects*. We then evaluated all models under these conditions, and the results are shown in Figure 4. It can be seen that for π_0 , π_0 -Fast, RIPT-VLA, UniVLA, and WorldVLA, the success rate decreases only marginally when confounding objects are added, indicating that these models, through training, indeed manage to focus their attention on the target objects. However, when the target objects’ placement is altered, the performance of the models drops significantly, suggesting that the current models may have merely learned the positional information of the target objects rather than truly capturing the high-level task-relevant semantics.

How Do Models Maintain Performance Under Illumination Changes?

We observe that for several models, the performance drop under light perturbations is limited to around 10 points, suggesting a surprising insensitivity

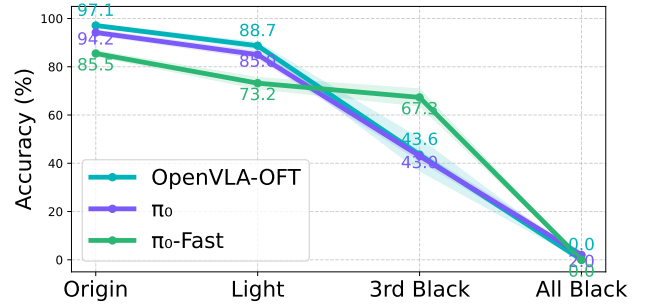


Figure 5. Illumination robustness and extreme ablation tests. The term *Light* denotes the condition with light perturbation applied. *3rd Black* and *All Black* represent conditions where only the third-view image is masked and where images from both views are masked, respectively.

to illumination changes. To investigate this phenomenon, we design an extreme ablation test: (i) *all-black*, where all camera inputs are replaced with black frames, and (ii) *3rd-black*, where only the third-person view is masked while the wrist camera is preserved. In the *all-black* condition, performance collapses to nearly zero across models, confirming a strong reliance on visual input. In contrast, under the *3rd-black* setting, the same models still achieve accuracies of 43.6, 43.0, and 67.3, respectively, demonstrating that the wrist view alone provides critical and stable close-range ge-

ometric and contact cues. This explains why standard light perturbations cause only minor degradation: illumination changes primarily affect the third-person view and global appearance, whereas the wrist view remains relatively stable. Consistently, models such as OpenVLA, Nora, and WorldVLA—which depend exclusively on third-person observations—suffer severe drops under light perturbations (often exceeding 60 points).

Based on our deeper investigation into object layout and illumination robustness, we can conclude:

Finding 5: Current VLAs exhibit positional bias rather than genuine semantic understanding of objects. While models demonstrate an ability to ignore distracting objects, they fail to generalize when target objects are displaced, indicating that they rely on memorized positional cues rather than learning invariant object semantics.

Finding 6: Wrist cameras provide critical robustness to illumination changes. The relative stability of performance under light perturbations is largely attributable to the wrist camera’s close-range perspective, which provides illumination-invariant geometric cues. Models lacking wrist-camera inputs show significantly greater vulnerability to lighting variations.

6. Do Contemporary VLA Models Truly Follow Language Instructions?

In the experiments presented in Section 4, we observed an intriguing phenomenon: when introducing language perturbations, the overall performance of the OpenVLA-OFT model was barely affected and remained close to the baseline level. To further investigate the potential underlying reasons, we propose the following three hypotheses:

(1) *The model may possess strong generalization capabilities in the language domain*, allowing it to remain robust even when instructions are perturbed.

(2) *The model may extract limited keywords from the input instruction for matching and decision-making*, rather than genuinely understanding the full semantic structure. However, this is unlikely because our perturbations include a *commonsense* subclass that performs keyword commonsense rewrite, yet the performance drop remains nearly negligible.

(3) *The model may not fully utilize the language modality, instead relying primarily on visual or other non-linguistic signals to complete tasks*. In such a scenario, language inputs would be functionally redundant, and even significant perturbations would have minimal impact.

To verify which of the above hypotheses is more plausible, we conducted additional analysis experiments.

6.1. What If We Remove Language, Does Performance Drop?

We introduced a **blank instruction** experiment. In this setting, the language input provided to the model was entirely replaced with an empty value, i.e., no linguistic information was supplied during inference. This approach directly tests whether the absence of language leads to a substantial performance degradation. We conducted experiments on all four suites of LIBERO, and the results are shown in Figure 6(a).

Surprisingly, even without any valid language input, the performance of OpenVLA-OFT on the *object* suite remained largely unchanged, with significant degradation observed only on the *long* suite. We attribute this to the greater reliance on instruction guidance in long-horizon tasks, which forces the model to attend to the language modality. This finding is highly revealing: although the model is nominally designed as a Vision-Language-Action (VLA) framework, in practice it degenerates into a form that disregards language, behaving more like a Vision-Action (VA) model.

6.2. What If We Replace Goals with OOD Objects, Do Models Fail?

We further designed a **goal replacement** task to directly examine whether models genuinely possess language instruction-following ability. Specifically, for the *layout* suite, where the issue appeared most pronounced, we replaced the target object in the instruction and the task goal with alternatives within the same scene. As shown in Figure 6(b), the experimental results revealed two key findings:

Finding 7: VLA models do not possess strong cross-object instruction-following generalization. In tasks with replaced targets, the model’s success rate dropped nearly to zero, with the degradation particularly severe for OpenVLA-OFT. The apparent “robustness” observed in prior language perturbation experiments did not stem from a deep modeling of language but rather from ignoring linguistic inputs altogether, leading to a superficially stable performance under perturbations.

Finding 8: VLA models appear to rely more on fixed vision-action mappings than on fully exploiting language signals in task decision-making. By analyzing roll-out cases, we observed that even when the target in the instruction was explicitly changed, the model still tended to execute the original target action rather than adjust its behavior according to the new instruction. More details can be found in Appendix E.

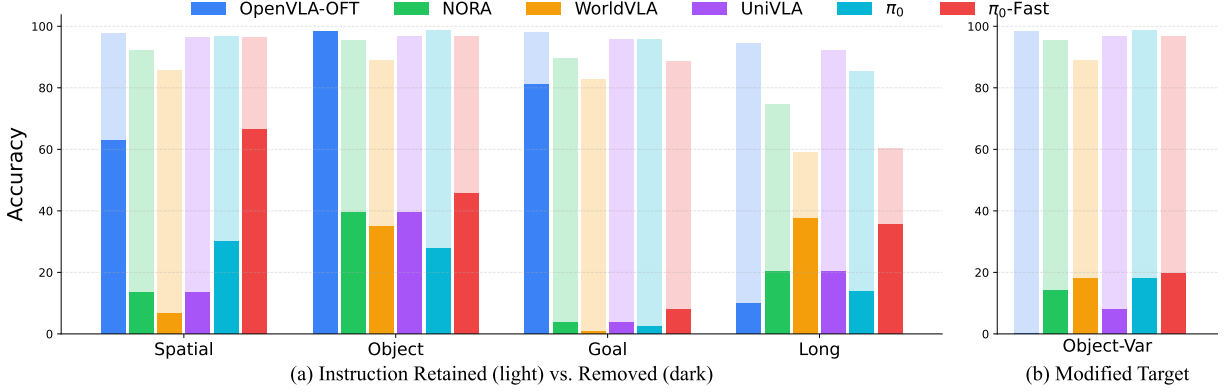


Figure 6. Accuracy of different models on instruction removed (a) and target modified (b) tasks. Light bars: original success rate with language instruction; (a) dark bars: success rate after removing the instruction; (b) Dark bars: success rate under altered task goal and instruction (task substitution).

7. Does There Exist Compositional Generalization Gap Across Multi-Dimensional Perturbations?

Generalization results under single-dimension perturbations demonstrate the model’s robustness against isolated factors. However, these dimensions may not be independent, and different types of perturbations are likely to exhibit complex dependencies. In this study, we refer to such performance as *compositional generalization*. To ensure scientific rigor, we define the problem from a statistical perspective as follows.

7.1. Statistical Definition of the Compositional Generalization Gap

We define the random variables D_i as

$$D_i = \begin{cases} 1, & \text{if the } i\text{-th type of perturbation is applied,} \\ 0, & \text{otherwise,} \end{cases}$$

and similarly for D_j . For a single trial, we define the success indicator variable

$$Y = \begin{cases} 1, & \text{if the task is successfully executed,} \\ 0, & \text{otherwise.} \end{cases}$$

The success rate can be defined in terms of conditional probability as $(D_i = d_i, D_j = d_j) = P(Y = 1 \mid D_i = d_i, D_j = d_j)$, $d_i, d_j \in \{0, 1\}$. We further estimate the joint probability between D_i and D_j conditioned on $Y = 1$,

$$p(D_i = d_i, D_j = d_j \mid Y = 1) = \frac{s(D_i = d_i, D_j = d_j)}{\sum_{a,b \in \{0,1\}} s(D_i = a, D_j = b)}$$

which represents the probability that the combination $D_i = d_i$ and $D_j = d_j$ occurs among all successful cases. Similarly, the marginal probabilities are

$$\begin{aligned} p(D_i = 1 \mid Y = 1) &= p(D_i = 1, D_j = 0 \mid Y = 1) + p(D_i = 1, D_j = 1 \mid Y = 1) \\ &= \frac{s(D_i = 1, D_j = 0) + s(D_i = 1, D_j = 1)}{\sum_{a,b \in \{0,1\}} s(D_i = a, D_j = b)}, \end{aligned} \quad (1)$$

$$\begin{aligned} p(D_j = 1 \mid Y = 1) &= p(D_i = 0, D_j = 1 \mid Y = 1) + p(D_i = 1, D_j = 1 \mid Y = 1) \\ &= \frac{s(D_i = 0, D_j = 1) + s(D_i = 1, D_j = 1)}{\sum_{a,b \in \{0,1\}} s(D_i = a, D_j = b)}. \end{aligned}$$

Intuitively, $p(D_i = 1 \mid Y = 1)$ reflects the probability that the i -th perturbation occurs among all successful cases. It measures the contribution of the i -th perturbation to the overall successful outcomes. A high value indicates that the perturbation frequently co-occurs with successful trials, suggesting the model is robust to this perturbation, while a low value indicates sensitivity to this perturbation. Similarly,

$$p(D_i = 1, D_j = 1 \mid Y = 1) = \frac{s(D_i = 1, D_j = 1)}{\sum_{a,b \in \{0,1\}} s(D_i = a, D_j = b)}$$

represents the proportion of successful cases under the “double perturbation” scenario. A high probability suggests the model maintains performance under joint perturbations, whereas a low probability indicates that the combination severely affects success.

In this study, we focus on the *Compositionality Gap* which is also the covariance between variable D_i and D_j given that $Y = 1$:

$$\begin{aligned} \Delta_{ij} &\triangleq \text{Cov}(D_i, D_j \mid Y = 1) \\ &= \mathbb{E}[D_i D_j \mid Y = 1] - \mathbb{E}[D_i \mid Y = 1] \mathbb{E}[D_j \mid Y = 1] \\ &= p(D_i = 1, D_j = 1 \mid Y = 1) \\ &\quad - p(D_i = 1 \mid Y = 1) p(D_j = 1 \mid Y = 1). \end{aligned}$$

The sign of Δ_{ij} correctly reflects the correlation of the contributions of the two perturbations to successful outcomes. Specifically: $\Delta_{ij} > 0$ indicates positive correlation, meaning the model can jointly handle both perturbations. $\Delta_{ij} < 0$ indicates negative interaction, meaning that the combination introduces additional difficulty beyond independent effects. $\Delta_{ij} = 0$ indicates no interaction, satisfying the independence assumption.

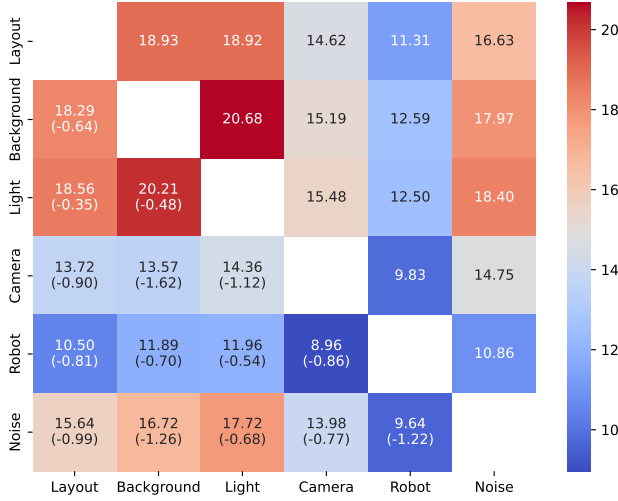


Figure 7. Heatmap of conditional probabilities under pairwise perturbations. Upper triangular entries represent independence-based products of single-dimension probabilities, while lower triangular entries show actual joint outcomes.

7.2. Experimental Setup and Results Analysis

We perform a total of 30k independent repeated experiments across 6 different perturbation (excluding language) combinations using the OpenVLA-OFT model to ensure high statistical significance. We perform independent tests for each type of single-dimension perturbation and pairwise perturbations, with 2000 trials conducted for each dimension combination.

The final experimental results are presented in a heatmap shown in Figure 7. The values in the upper-triangular matrix A_{ij} ($1 \leq i < j \leq 6$) are the product of the conditional probabilities of two single-dimension perturbations. The values in the lower-triangular matrix A_{ij} ($1 \leq j < i \leq 6$) represent the actual probabilities when applying joint perturbations. Additionally, we calculate the compositional generalization gap $\Delta_{ij} = A_{ij} - A_{ji}$ ($1 \leq j < i \leq 6$) and verify the statistical significance of the results using a chi-squared test, as shown in Appendix F.

Finding 9: Generalization is intrinsically non-decomposable. The consistent negative compositionality gap reflects interaction effects among perturbations, where co-occurring shifts act as coupled noise sources in feature space and expose entanglement in the learned representations. The findings indicate that current models lack mechanisms to capture higher-order dependencies, leading to pronounced robustness degradation under complex perturbation combinations.

8. Does Training on Generalized Sets Improve Generalization?

Leveraging our highly automated generalization pipeline, we constructed an extensive training dataset comprising over 20,000 successful trajectories. This dataset was con-

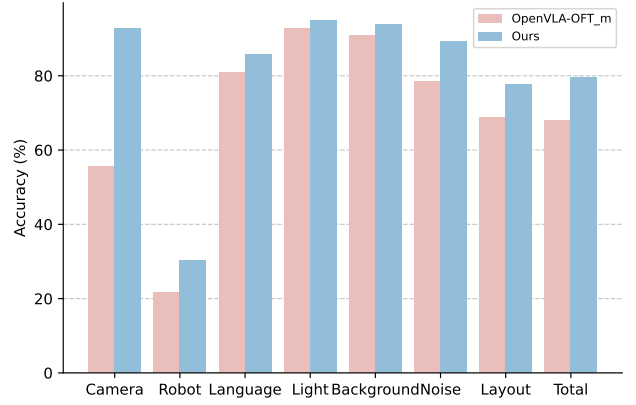


Figure 8. Performance comparison across different perturbation types after training on the augmented dataset.

structed through a substantial expansion of the original LIBERO benchmark, greatly increasing the number of trajectories and scene diversity, enabling a systematic evaluation of how generalization-oriented training affects model performance. Further details on dataset construction are available in Appendix D.

Using this dataset, we conducted mixed fine-tuning starting from the official OpenVLA-OFT weights. As shown in Figure 8, our method achieves the highest overall success rate (79.6%), outperforming all baseline models across nearly all perturbation types. Most notably, it exhibits a dramatic improvement in camera view robustness (92.8%), surpassing the next best model by 37.2 percentage points. Significant gains are also observed under noise (89.3%) and layout (77.6%) perturbations. These results demonstrate that training with our generalized dataset substantially enhances model robustness to a wide range of unseen environmental variations.

9. Conclusion

In this paper, we introduced LIBERO-Plus, a systematic framework for progressive robustness evaluation that provides comprehensive perturbation coverage across 7 perturbation, automated generation of over 56K robustness scenarios, and fine-grained analysis through progressive difficulty stratification. Evaluation of 10 popular VLA models reveals a significant generalization gap, contrasting their near-saturated performance on benchmarks like LIBERO. Most VLA models remain brittle, showing particular vulnerability to camera and robot state changes, almost all models ignore the language instructions, and some of the models execute with a bare memorization of the trajectory instead of relying on visual feedback. We also identify positional bias and negative combinatorial generalization gaps under combined perturbations. We urge the community to prioritize true task diversity in evaluation and develop architectures that generalize robustly beyond constrained benchmark settings.

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